

All conditions required:

1. Flat metric
2. Constraints Fourier-eliminable (operators $\in$ exact multipliers, no time-dependent coefficients)
3. All BCs periodic
4. All operators modal-supported (15 supported: identity, laplacian, $\text{gradient}_{x,y,z}$, $\text{laplacian}_{x,y,z}$, $\text{cross_deriv}_{xy,xz,yz}$, biharmonic, $\text{derivative}_3_{x,y,z}$)
5. No time-dependent coefficients

requires all

Solver Key

	Modal	Padé matrix-exp (path D) / <code>expm_multiply</code>	$\sim 10^{-16}$ precision
	IDA	SUNDIALS implicit DAE	$\sim 10^{-8}$ tolerance
	CVODE	SUNDIALS adaptive ODE	$\sim 10^{-8}$ tolerance
	Leapfrog	Symplectic KDK/Yoshida	Manual only
	scipy	DOP853/Radau/BDF	Manual only

Leapfrog and scipy are never auto-selected (require explicit solver specification).

Auto-select solver
scheme, spec, grid, bc

Explicit
scheme?

Yes

Use specified
solver

No (auto)

Modal
eligible?

Yes

Modal solver

No

Constraint
equations?

Yes

IDA

*zeroth-order in t
(elliptic / algebraic DAE)*

No

First-order
equations?

Yes

IDA

*first-order in t
(diffusion / transport)
(implicit BDF)*

No

Dissipation?

Yes

IDA

*dissipative ∂_t terms
(breaks symplecticity)*

No

Missing
Hamiltonian?

Yes

Warning $\rightarrow$
CVODE fallback

*no Hamiltonian available
for second-order wave eqs*

No

CVODE

*Pure wave, Hamiltonian
(adaptive BDF)*